lecture-06

$$(\mathbb{R}^2, +, \odot, \mathbb{R})$$ field

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$ ✓ (A1-A5: satisfied)

scalar, vector

$$\underbrace{R \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}}$$

$\underbrace{\text{M1-M5}}$

(M2) $\alpha(u+v) = \alpha u + \alpha v$

$$\alpha\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) = \alpha \odot \underbrace{\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\underbrace{\alpha \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}} + \underbrace{\alpha \odot \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

M2 ✓

(M4) $\underbrace{\alpha \odot (\beta \odot u)} = (\alpha * \beta) \odot u$

$$\alpha \odot \left(\beta \underbrace{\odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}\right) = \underbrace{\alpha \odot \begin{pmatrix} 0 \\ 0 \end{pmatrix}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\underbrace{(\alpha * \beta)} \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \underbrace{(\alpha\beta) \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

M4 ✓

(M5) $\underbrace{1_{\mathbb{F}}}_{\text{scalar}} \in \underbrace{\mathbb{F}}_{\text{field}} = \{\mathbb{R}, \mathbb{C}, \ldots\}$

$$1_{\mathbb{F}} \odot u = u \odot 1_{\mathbb{F}} = u$$

$1_{\mathbb{F}}$ (identity)

So, there we first change the multiplication rules $\alpha \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

hence, our $\boxed{\text{identity}}$ will not be change because our $\boxed{\text{field is still same}}$

1 $\qquad$ $\mathbb{R}$

$$1 \odot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \neq \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

Hence, M5 is not satisfied. $\Longrightarrow$ This is not a vector space.

Example

$$\underbrace{\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 + 1 \\ y_1 + y_2 + 1 \end{pmatrix}}$$

$$k \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} kx_1 \\ ky_1 \end{pmatrix}$$

(A3) $\boxed{(u+v)+w = u+(v+w)}$

$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ $\quad v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ $\quad w = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$

$$(u+v)+w = \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) + \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

$$= \begin{pmatrix} x_1+x_2+1 \\ y_1+y_2+1 \end{pmatrix} + \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

$$= \begin{pmatrix} x_1+x_2+1+x_3+1 \\ y_1+y_2+1+y_3+1 \end{pmatrix} = \begin{pmatrix} x_1+x_2+x_3+2 \\ y_1+y_2+y_3+2 \end{pmatrix}$$

$$u+(v+w) = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2+x_3+1 \\ y_2+y_3+1 \end{pmatrix} = \begin{pmatrix} x_1+x_2+x_3+1+1 \\ y_1+y_2+y_3+1+1 \end{pmatrix}$$

$$= \begin{pmatrix} x_1+x_2+x_3+2 \\ y_1+y_2+y_3+2 \end{pmatrix}$$

A4 Fix e such that,

$$\boxed{\underbrace{e+u} = u+e = \underbrace{u}}$$

$$e = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \qquad \underbrace{e+u} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha+x_1+1 \\ \beta+y_1+1 \end{pmatrix}}$$

Let's equal

$$\underbrace{e+u} = u$$

$$\Rightarrow \begin{pmatrix} \alpha+x_1+1 \\ \beta+y_1+1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\alpha+\cancel{x_1}+1 = \cancel{x_1} \Rightarrow \alpha = -1$$

$$\beta+\cancel{y_1}+1 = \cancel{y_1} \Rightarrow \beta = -1$$

$$e = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

(A5) For each u there exist an unique u' such that,

$$\boxed{u' + u = u + u' = e}$$

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \qquad u' = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

check

$$u' + u = e$$

$$\Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} \alpha + x_1 + 1 \\ \beta + y_1 + 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$\alpha = -2 - x_1$$

$$\beta = -2 - y_1$$

(M2) $\boxed{\alpha \cdot (u+v) = \alpha \cdot u + \alpha \cdot v}$

$$\alpha\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) = \alpha \begin{pmatrix} x_1 + x_2 + 1 \\ y_1 + y_2 + 1 \end{pmatrix} = \begin{pmatrix} \alpha(x_1 + x_2 + 1) \\ \alpha(y_1 + y_2 + 1) \end{pmatrix}$$

$$= \begin{pmatrix} \alpha x_1 + \alpha x_2 + \underline{\underline{\alpha}} \\ \alpha y_1 + \alpha y_2 + \underline{\underline{\alpha}} \end{pmatrix}$$

$$\underbrace{\alpha \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}} + \alpha \cdot \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \alpha x_1 \\ \alpha y_1 \end{pmatrix} + \begin{pmatrix} \alpha x_2 \\ \alpha y_2 \end{pmatrix} = \begin{pmatrix} \alpha x_1 + \alpha x_2 + \underline{\underline{1}} \\ \alpha y_1 + \alpha y_2 + \underline{\underline{1}} \end{pmatrix}$$

M2 is not satisfied..

Adding by a constant number

Hack

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 + k \\ y_1 + y_2 + k \end{pmatrix} \qquad (M2)$$

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 y_2 \end{pmatrix} \qquad (A5)$$

Example

$$1 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$1 \cdot u = u$$

Subspace

$$X = \{ \ldots \}$$

$$A = \{ \ldots \} \subseteq X$$

$$(\underbrace{X}_{\text{Set}}, +, \cdot, \mathbb{F})$$

$$(A, +, \cdot, \mathbb{F})$$

$$A \subset X \text{ (Subset)}$$

Example

$$S = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x, y \in \mathbb{R} \right\} \subset \mathbb{R}^3$$

subset

Verify S is a subspace or not?

① $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \in S \Rightarrow S \neq \emptyset$

fix

②, $u, v \in S \qquad u = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} x_2 \\ y_2 \\ 0 \end{pmatrix}$

$$u + v = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ 0 \end{pmatrix} \in S$$

fix

③ $\alpha \in \mathbb{R}, \quad u \in S, \quad u = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix}$

$$\alpha \cdot u = \alpha \cdot \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha x_1 \\ \alpha y_1 \\ 0 \end{pmatrix} \in S$$

fix

S is a subspace of $\mathbb{R}^3$.

$$S = \left\{ \begin{pmatrix} x \\ y \\ \cancel{0} \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x, y \in \mathbb{R} \right\}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \qquad \mathbb{R}^2$$

Example:

$$T = \left\{ \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x, y \in \mathbb{R} \right\}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ 1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ 1 \end{pmatrix} = \boxed{\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ 2 \end{pmatrix}} \notin T$$

Example

$$W = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \;\middle|\; x + y = 10 \right\}$$

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \in W \qquad v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \in W$$

$$x_1 + y_1 = 10 \qquad\qquad x_2 + y_2 = 10$$

$$u + v = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \boxed{\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}} \notin W$$

$$(x_1 + x_2) + (y_1 + y_2) = (x_1 + y_1) + (x_2 + y_2) = 10 + 10 = 20 \neq 10$$